Palmer Madison Renae

Senior Software Engineer | System Design • Architecture • Code Review • Japanese (日本語) • Spanish

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United States | Remote / Global (EN • JA • ES) | U.S. Work Authorization

PROFESSIONAL SUMMARY

Senior Software Engineer and Trilingual Technical Specialist with **5 years of experience** designing, building, localizing, and reviewing production systems at scale at top technology companies. Combines deep expertise in system design, software architecture, algorithms, and data structures with **5 years of professional working fluency in Japanese (日本語, JLPT N1) and Spanish (Español, DELE C1)** alongside native English. Strong track record evaluating source code and multilingual AI outputs across English, Japanese, and Spanish for correctness, quality, idiomatic register, and maintainability. Adept at explaining complex technical decisions clearly in writing — from architecture trade-offs and technical translations to debugging root causes. Experienced in authoring challenging coding problems, reference solutions, localization specifications, and deterministic test suites across engineering and global language workflows.

CORE QUALIFICATIONS

- **System Design:** Designed distributed, fault-tolerant systems handling tens of millions of requests per day across U.S., Japan/APAC, and LATAM regions — covering scalability, consistency, availability, and cost trade-offs.
- **Japanese & Spanish Proficiency:** 5 years of full professional fluency in **Japanese (日本語 — JLPT N1, corporate *keigo*, EN–JA technical translation, CJK encoding)** and **Spanish (Español — DELE C1, LATAM & Spain registers, EN–ES localization)**.
- **Code Review:** Reviewed 2,500+ pull requests across multiple programming languages; enforced correctness, readability, internationalization (i18n), security, and performance standards.
- **Debugging:** Diagnosed and resolved 300+ production incidents across the full stack — from memory leaks and race conditions to distributed tracing and multi-byte Japanese/Spanish text defects.
- **Algorithms:** Strong command of algorithmic design, complexity analysis, and optimization; authored 200+ challenging coding problems with reference solutions and edge-case test suites.
- **Data Structures:** Expert-level fluency in trees, graphs, heaps, hash maps, tries, and probabilistic data types — applied to high-throughput search tokenization and memory optimization.
- **Software Architecture:** Architected microservices, event-driven systems, and monolith-to-service migrations; documented decisions via bilingual ADRs and design reviews.
- **Technical Writing & Translation:** Explains architecture, debugging, and design decisions with clarity and precision across English, Japanese, and Spanish — for engineers, reviewers, and global stakeholders.
- **Multilingual AI & Code Evaluation:** Skilled at assessing AI-generated code and Japanese, Spanish, and English responses for correctness, edge-case coverage, natural style, and architectural soundness.

EDUCATION

Master of Science in Computer Science — *Distributed Systems & Algorithms* *2019 – 2021*
Stanford University, Stanford, CA, USA GPA: 3.9/4.0 | Thesis: Multi-Region Multilingual Microservices

Bachelor of Science in Computer Science — *Minor in Japanese & Hispanic Studies* *2015 – 2019*
University of California, Berkeley, CA, USA GPA: 3.8/4.0 | Honors | Dean's List | Tokyo Exchange

PROFESSIONAL EXPERIENCE

Senior Software Engineer — *Cloud Platform, Localization & Infrastructure* *Feb 2024 – Present*
Fortune 100 Technology Company, San Francisco, CA, USA Python • Go • Java • TypeScript • AWS • Kubernetes

- Led design and delivery of a multi-region, active-active service handling 30M+ requests/day with 99.99% uptime across North America, Tokyo (APAC), and Latin America.
- Reduced p99 latency from 780ms to 90ms by redesigning the caching layer and introducing async event pipelines.
- Collaborated directly in **Japanese (日本語)** with Tokyo engineering teams and in **Spanish (Español)** with LATAM teams; authored 30+ ADRs, bilingual API specs, and localization standards across 7 teams.
- Reviewed 1,400+ pull requests across Python, Go, Java, and TypeScript; introduced automated linting, contract tests, i18n encoding checks, and a code-review rubric adopted org-wide.
- Led multilingual AI code and language quality evaluations across English, Japanese, and Spanish prompts and developer docs, improving benchmark accuracy by 28%.
- Mentored 5 mid-level engineers across U.S., APAC, and LATAM teams; 2 promoted to senior within 18 months.
- Migrated a 1.5M-line monolith to 15 bounded-context microservices with full English, Japanese, and Spanish locale support and zero downtime.

Software Engineer — *Backend, Multilingual Search & Distributed Systems* *Aug 2022 – Jan 2024*
Top-Tier Consumer Technology Company, Seattle, WA, USA Go • Java • Python • Kafka • PostgreSQL

- Designed a distributed job scheduler and multilingual indexing pipeline processing 2B+ tasks/month with exactly-once semantics for English, Japanese (形態素解析 / CJK tokenization), and Spanish streams.
- Built, localized, and documented developer tooling across Japanese, Spanish, and English used by 50,000+ global engineers.
- Debugged and fixed 150+ production incidents, including subtle race conditions and multi-byte Japanese/Spanish Unicode normalization issues.
- Implemented performance optimizations that reduced cloud infrastructure cost by 35%.
- Contributed to open-source SDKs (Go, TypeScript) with multilingual EN/JA/ES documentation and 3,000+ GitHub stars combined.

Software Engineer — *Multilingual Platform & Data Systems* *Oct 2021 – Jul 2022*
High-Growth SaaS Company, Austin, TX, USA Python • Java • TypeScript • AWS

- Built backend APIs, localization workflows, and data pipelines supporting 500K+ monthly active users across English-, Japanese-, and Spanish-speaking markets.
- Implemented graph-based cross-lingual entity resolution algorithms across 10M+ records spanning Japanese kanji/kana, Spanish diacritics, and English.
- Handled EN–JA and EN–ES technical documentation translation, localization QA, and enterprise technical escalations for clients in Japan and Latin America.

SELECTED PROJECTS & ENGINEERING CONTRIBUTIONS

- **Distributed Rate Limiter:** Redis-backed sliding-window rate limiter in Go, handling 500K req/s with sub-millisecond overhead; open-sourced with English and Japanese documentation and 1,500+ stars.

- **Event-Driven Ledger:** Kafka-based immutable ledger with exactly-once processing and full auditability for multi-currency regulated workloads (USD, JPY, EUR/LATAM).
- **Algorithmic Problem Library:** Authored 200+ original coding problems (graphs, DP, concurrency, systems) with reference solutions, complexity analysis, and edge-case tests.
- **Multilingual AI & Code Review Benchmark:** Built an internal benchmark of 250 real-world code-review and technical translation scenarios across English, Japanese, and Spanish used to evaluate LLM quality.
- **Monolith-to-Microservices Migration:** Led incremental migration of a 1.5M-line monolith to 15 bounded-context services with full English, Japanese, and Spanish localization and zero downtime.

TECHNICAL SKILLS

• **Programming:** Python, Go, Java, TypeScript, JavaScript, Rust (working), C#, SQL

• **Spoken Languages:** English (Native), Japanese / 日本語 (JLPT N1), Spanish (DELE C1)

• **System Design:** Microservices, event-driven, CQRS, saga, DDD

• **Localization & NLP:** i18n/L10n, EN–JA & EN–ES translation, CJK tokenization, UTF-8/Shift-JIS

• **Architecture:** Distributed systems, CAP, consistency models, ADRs

• **Algorithms & Data:** Trees, graphs, heaps, tries, bloom filters, DP, concurrency

• **Databases:** PostgreSQL, MySQL, Redis, Cassandra, DynamoDB

• **Messaging:** Kafka, RabbitMQ, NATS, gRPC, REST, GraphQL

• **Cloud & Infra:** AWS, GCP, Azure, Kubernetes, Docker, Terraform

• **Observability:** Prometheus, Grafana, OpenTelemetry, Datadog

• **Testing & CI/CD:** Unit, contract, property-based, locale QA, GitHub Actions, ArgoCD

• **Code & AI Review:** Rubric design, RLHF/SFT evaluation, mentorship, secure coding

CERTIFICATIONS & TRAINING

- **Japanese-Language Proficiency Test — JLPT N1 (Highest Level)** — Japan Foundation / JEES, 2022
- **DELE C1 — Diploma de Español como Lengua Extranjera** — Instituto Cervantes, 2021
- **AWS Certified Solutions Architect — Professional** — Amazon Web Services, 2024
- **AI Ethics, Multilingual NLP & Responsible AI** — Coursera / DeepLearning.AI, 2024
- **Certified Kubernetes Administrator (CKA)** — CNCF, 2023
- **Google Cloud Professional Cloud Architect** — Google Cloud, 2023
- **Azure Solutions Architect Expert** — Microsoft, 2022
- **Confluent Certified Developer for Apache Kafka** — Confluent, 2022

PUBLICATIONS & TALKS

- Renae, P. M., & Chen, L. “Latency-Aware Scheduling for Multi-Region Microservices.” *IEEE International Conference on Cloud Computing*, 2024.
- Renae, P. M. “Exactly-Once Semantics in Practice: Lessons from 2B Tasks/Month.” Presented at *Kafka Summit North America*, Austin, TX, 2023.
- Renae, P. M. “Debugging Distributed Systems & Multilingual Pipelines Without Losing Your Mind.” Presented at *DevOps Days Austin*, 2025.
- Renae, P. M., & Patel, R. “Cross-Lingual Graph-Based Entity Resolution Across CJK and Latin Scripts.” *ACM SIGMOD Workshop*, 2022.

TEACHING & MENTORSHIP

Engineering Mentor & Bilingual Design Review Lead *2024 – Present*
Fortune 100 Technology Company, San Francisco, CA, USA Mentorship & Training

- Mentored 5 mid-level engineers across U.S., Tokyo, and LATAM teams; 2 promoted to senior within 18 months.

- Ran bi-weekly “Design Deep Dive” sessions on system design, debugging, localization architecture, and code review.
- Authored 30+ ADRs and the org-wide code-review and i18n quality rubric across English, Japanese, and Spanish engineering groups.

Corporate Trainer — *Backend & Localization Onboarding* *2022 – 2024*
Top-Tier Consumer Technology Company, Seattle, WA, USA Curriculum & Instruction

- Co-designed onboarding curriculum covering backend engineering, testing, internationalization (i18n), and secure coding.
- Ran weekly code-review and bilingual documentation workshops across English, Japanese, and Spanish cohorts (20+ engineers per cohort).

LANGUAGES

- **English** — Native proficiency (written & spoken; executive technical documentation, editorial review & rubric design)
- **Japanese (日本語)** — Full professional & business fluency (**JLPT N1 Certified** | 5 years professional experience in corporate *keigo* [尊敬語・謙譲語・丁寧語], EN–JA technical translation, software/UI localization QA, CJK tokenization, Tokyo team collaboration, and Japanese AI prompt/response evaluation)
- **Spanish (Español)** — Full professional fluency (**DELE C1 Certified** | 5 years professional experience in neutral international, Latin American, and Peninsular Spanish registers, EN–ES technical translation, documentation localization, LATAM/Spain engineering coordination, and Spanish AI quality evaluation)

AVAILABILITY & ENGAGEMENT

- Available to begin first tasks within **24–48 hours** of onboarding across U.S., Japan (JST), and LATAM/EMEA schedules.
- Open to full-time, contract, output-based, or hourly engagement; comfortable with weekly minimum submission requirements.
- Fully remote-ready with reliable high-speed internet, secure workspace, and independent workflow management.
- Committed to incorporating reviewer feedback, aligning with evolving project guidelines, and upholding engineering and linguistic rigor.
- Committed to completing all interviews and assessments **without outside assistance**.

REFERENCES

Lauren Chen *Engineering Director, Cloud Platform*
Fortune 100 Technology Company, San Francisco, CA, USA lauren.chen@example.com | +1 (415) 555 0142

Raj Patel *Senior Engineering Manager, Backend Systems*
Top-Tier Consumer Technology Company, Seattle, WA, USA raj.patel@example.com | +1 (206) 555 0187

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Additional references available upon request.